Supply Chain Intelligence: Shipment Performance and Delivery Risk Analytics

Executive summary | DataCo Smart Supply Chain dataset | Python, MySQL, Power BI | Arghadeep Ghosh, Data Analytics Consultant

Objective

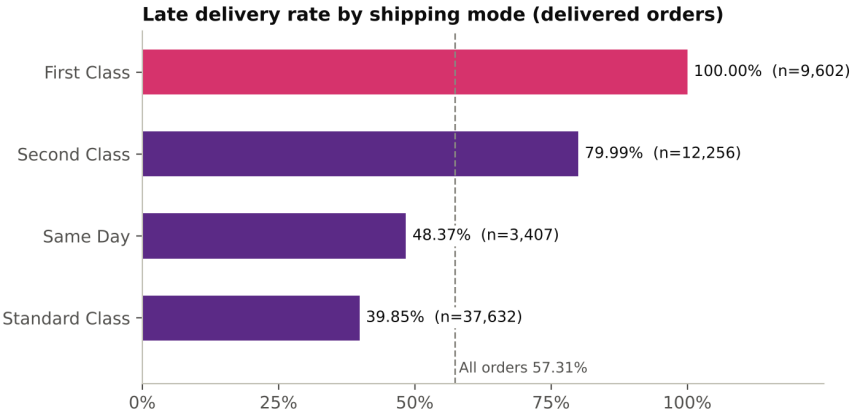
Find out why 57% of delivered orders arrive after the promised date, which parts of the operation are responsible, and what the business could change. The analysis covers 180,519 order-item records (65,752 orders, January 2015 to January 2018), measured at order level because a delivery is an order-level event.

Tools and method

- **Python (pandas):** loaded and validated the data, removed 13 empty, personal or duplicate columns, translated country names, engineered delay and scope flags, and flagged 7,754 canceled or suspected-fraud rows instead of deleting them so delivery KPIs use delivered orders only.
- **MySQL 8:** answered 25 business questions with aggregation, CTEs, window functions and a what-if scenario query, cross-checked against the Python results.
- **Power BI:** one-page dashboard with KPI cards, shipping-mode, promised-versus-actual and monthly-trend visuals, and market and segment slicers.

Key results (62,897 delivered orders)

Measure	Result
Late delivery rate	**57.31%** (36,048 orders); late orders average 1.62 days late
Promised vs actual days	2.94 promised vs 3.50 actual on average
Late rate by shipping mode	First Class **100.00%**, Second Class 79.99%, Same Day 48.37%, Standard Class 39.85%
Concentration	First + Second Class: 34.8% of orders, **53.8% of late orders**
Market / segment / month	56.55-57.68% / 57.08-57.59% / 55.04-59.86%: no statistically significant difference (chi-square p = 0.10 to 0.97)
Revenue and profit	31.64M net revenue, 3.81M profit (12.03% margin)
Canceled and fraud orders	1.41M, or 4.26% of net revenue, kept out of delivery KPIs



Late rate by shipping mode. First Class promises 1 day and always takes 2 days.

Main findings

- **Shipping mode is the dominant factor** (Cramer's V 0.48). Region, market, segment and month show no significant effect.
- **First Class is late every time:** 1-day promise, 2-day delivery, in all 23 regions and all four years.

- **Averages hide the risk:** Standard Class averages 3.99 days against a 4-day promise yet 39.85% of its orders are late, because delivery time is spread evenly from 2 to 6 days.
- **Lateness shows no measurable profit effect:** margin is 11.83% for late and 12.29% for on-time orders.

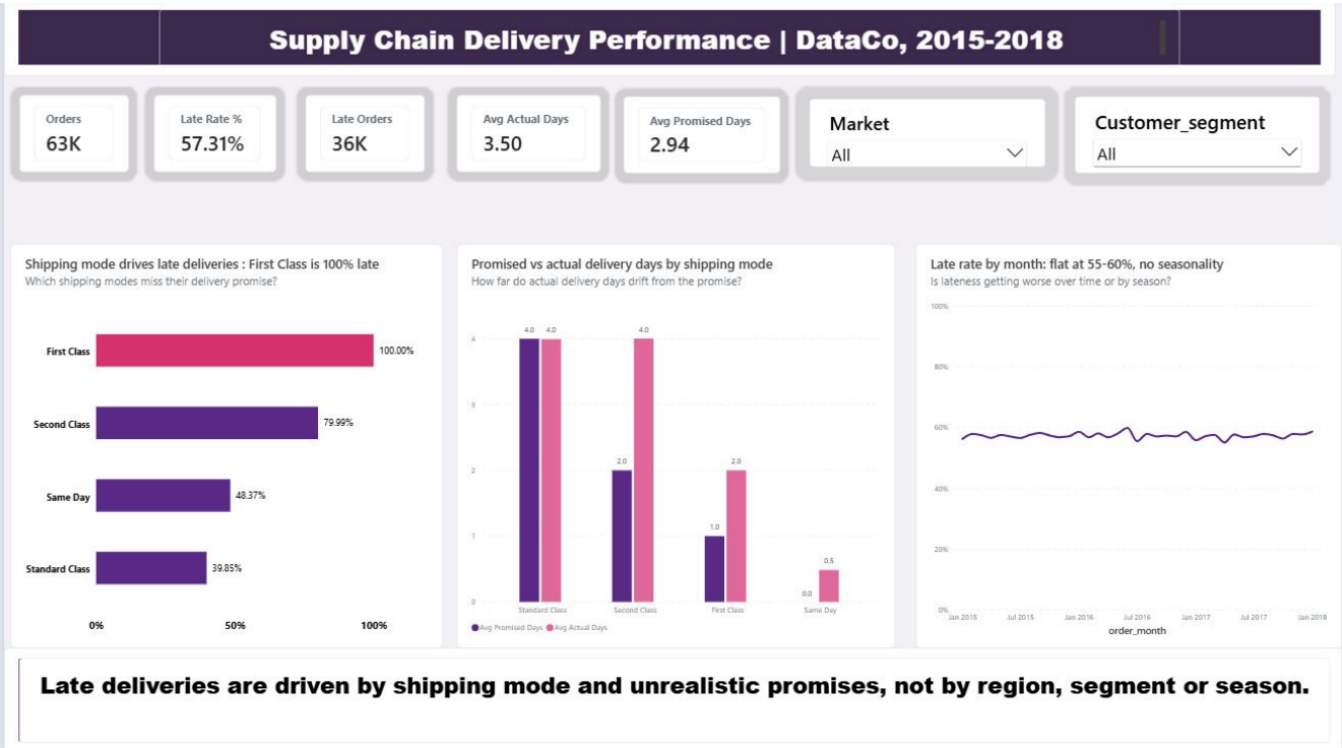
Business problems and recommended actions

Recommendations only; none was implemented. Effects come from a SQL what-if simulation that assumes transit times stay the same.

Problem	Recommended action	Modelled effect
Unrealistic First and Second Class promises	Re-base First Class to 2 days and Second Class to 4 days	Late rate 57.31% to **34.26%**
Wide transit spread (19,903 orders take 5-6 days)	Promise about the 80th percentile per mode, and shorten the slow tail	Late rate **15.89%**
No promise-accuracy tracking	Monthly KPI: late rate and actual-minus-promised days by mode, with an alert	Earlier detection of drift
Canceled and fraud orders inside delivery KPIs	Report them separately as a loss metric (1.41M)	Cleaner KPIs

Trade-off: longer promised delivery times may make the service less competitive. This should be tested with customers before any change.

Dashboard



Power BI dashboard: KPI cards, late rate by mode, promised vs actual days, monthly trend, market and segment slicers.

Business impact and next steps

The findings show that delivery promises, not geography or season, are the main lever on lateness in this dataset. Fixing First and Second Class promises is the highest-value step, and the dashboard can track promise accuracy monthly. Next steps: carrier and route data to explain slow transit, a customer test for longer promises, and an order-level model that predicts delivery dates at order time (using only information known at order time, not the delivery outcome).

Integrity note: the data appears simulated (uniform delivery days, almost no regional or category effect), so the results demonstrate the method and describe this dataset rather than real-world performance. The analysis is descriptive and shows association, not causation.